



Long-term effectiveness of bisoprolol in patients with angina: A real-world evidence study

M. Sabidó^{a,*}, Thilo Hohenberger^a, Guido Grassi^{b,c}

^a Merck KGaA, Frankfurter Strasse 250, Darmstadt, 64293, Germany

^b Clinica Medica, University Milano Bicocca, Milan, Italy

^c IRCCS Multimedica, Sesto San Giovanni, Milan, Italy

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ABSTRACT

A cohort analysis using UK Clinical Practice Research Datalink (CPRD) was performed to compare the effects of bisoprolol, other β -blockers, and drugs other than β -blockers on the long-term risk of mortality and cardiovascular events in patients with angina. Adult patients first diagnosed with angina from 2000 to 2014, with ≥ 365 days of registration to first angina diagnosis and initiating monotherapies of bisoprolol, other β -blockers, or drugs other than β -blockers within 6 months of angina diagnosis were included. Incidence rates for each treatment cohort were compared using adjusted hazard ratio (HR) and 95% confidence intervals (CI) obtained from Cox regression analyses. Overall, 987 patients were treated with bisoprolol, 1348 with other β -blockers and 5272 with drugs other than β -blockers. Over the total follow-up (≤ 14 years), the HR of bisoprolol versus other β -blockers and drugs other than β -blockers for mortality was 0.45 (95% CI: 0.34–0.61) and 0.50 (95% CI: 0.38–0.66), respectively. The HR of bisoprolol versus other β -blockers for angina was 0.58 (95% CI: 0.50–0.68) and versus drugs other than β -blockers was 0.77 (95% CI: 0.68–0.88), respectively. For myocardial infarction, the HR of bisoprolol versus drugs other than β -blockers up to 14 years was 0.34 (95% CI: 0.23–0.52) and versus other β -blockers up to 5 years was 0.45 (95% CI: 0.27–0.75). At 5 years, the HR of bisoprolol versus other β -blockers, and drugs other than β -blockers, for arrhythmia was 0.60 (95% CI: 0.35–1.0) and 0.61 (95% CI: 0.40–0.93), respectively. In conclusion, long-term significant reduction in the risk of mortality and various cardiovascular events with bisoprolol versus other β -blockers, and drugs other than β -blockers, confirm treatment guidelines recommendation that bisoprolol is particularly well suited as the first-line treatment of angina in primary care.

1. Introduction

Angina pectoris refers to the chest discomfort attributable to myocardial ischemia [1]. The most common cause of myocardial ischemia is atherosclerotic obstruction of one or more of the coronary arteries (coronary artery disease [CAD]) [1,2]. However, any significant imbalance between myocardial oxygen supply and demand can lead to myocardial ischemia and angina [2]. The prevalence of angina varies between countries [3]. In the UK the prevalence across all ages is reported to be 3.1% in men and 1.8% in women [4].

For many patients, the development of angina is the first clinical indication of CAD and is a signal that patients will need to initiate treatment to alleviate symptoms and reduce their risk of an acute event [5]. European and US guidelines recommend the use of β -blockers as first-line therapy for patients with angina [5–8]. In the UK, National

Institute for Health and Care Excellence guidelines advise that stable angina should be routinely treated with either a β -blocker or calcium channel blocker (CCB) [5]. Other drugs, such as nitrates, should be considered when β -blocker or CCB treatment has been clinically excluded as a treatment option [5,7]. These recommendations are based on several clinical studies that have reported the beneficial effects of using β -blockers to treat patients with angina [5,8–11]. Studies have demonstrated a reduction in ischemic/anginal episodes with β -blockers compared with placebo [5] and have indicated that there is a greater reduction in these episodes with β -blockers than with CCBs [11]. Meta-analyses have also provided evidence that β -blockers significantly reduce the risk of mortality in patients with angina compared with no treatment [9] and are associated with a lower rate of adverse events compared with CCBs [8].

As a class, β -blockers are competitive antagonists and function

* Corresponding author.

E-mail address: meritxell.sabido-espino@merckgroup.com (M. Sabidó).

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through diminishing the effect of the sympathetic nervous system which mediates the fight-or-flight responses, by blocking the receptor sites for the endogenous catecholamines epinephrine (adrenaline) and norepinephrine (noradrenaline) on adrenergic β -receptors [12,13]. One of the major actions of β -blockers is to decrease heart rate, cardiac contractility, and blood pressure (BP) [1]. However, this is a heterogeneous group of drugs, which differ in their selectivity for specific β -receptors, their intrinsic sympathomimetic activity, and their pharmacokinetic properties [1,14]. Bisoprolol is a β_1 -selective blocker and is indicated for the control of symptoms in patients with angina [15]. Studies have shown that bisoprolol can reduce the ischemic burden in patients with angina relative to the CCB nifedipine [11] and reduce angina attacks and improve exercise tolerance relative to isosorbide dinitrate [16].

Although clinical studies have shown that β -blockers are effective in the treatment of angina, there is a lack of head-to-head studies comparing individual β -blockers. In addition, many comparative studies are limited by a small sample size and short- to medium-term follow-up, so little is known about the impact of these agents on the long-term prognosis in patients with angina.

The objective of this study was to compare the incidence and risk of mortality and several cardiovascular (CV) events, including myocardial infarction (MI), angina, embolism, arrhythmia, or stroke in patients with angina receiving monotherapy with bisoprolol versus other β -blockers or drugs other than β -blockers in patients with angina in a real-world primary care setting.

2. Material and methods

2.1. Data source

Data were obtained from the UK Clinical Practice Research Datalink (CPRD) database from 2000 to 2014. The CPRD is a British National Health Service observational data and interventional research service. It is the main data source for patients at the primary care level in the UK collected from general practitioners [17,18]. CPRD includes data from 800 practices, accounting for 22 million total patients, and over 5 million active patients (alive and currently registered) [19]. This covers approximately 8.5% of the UK population with similar age, sex, and ethnicity distribution to the general population [20,21]. Follow-up duration for active patients ranges from 3.4 to 13.9 years (median: 9.4 years) [18]. The service provides universal coverage, and no segment of the population is excluded.

All patient data in the CPRD are de-identified to protect the privacy of patients and providers. The study was reviewed and approved by the CPRD Independent Scientific Advisory Committee for the Medicines and Healthcare Products Regulatory Agency database research.

2.2. Study population

The analysis included all patients in the CPRD database aged ≥ 18 years who were newly diagnosed with angina pectoris. Angina diagnosis was based on database READ codes and its first diagnosis was defined as the index date. Patients were included if there was no angina code in the patient's record in the pre-index period (ie, at least 365 days prior to the first angina diagnosis). In addition, adult patients were included if they were newly prescribed bisoprolol, another β -blocker, or another drug as monotherapy within 6 months of the first diagnosis of angina. Patients were excluded if they initiated combination therapy or had no treatment recorded after the first angina diagnosis, or received a β -blocker (bisoprolol or other β -blockers) prior to the index diagnosis. Each patient was followed from first recorded angina event until the date of patient death, last data collected, or treatment cohort switch, whichever came first.

2.3. Exposures

According to initial treatment recorded within 6 months of the index date, patients were categorized into three cohorts of patients receiving monotherapy. The bisoprolol cohort included bisoprolol fumarate. In the other β -blocker cohort, treatment with any other β -blocker except bisoprolol (acebutolol hydrochloride, atenolol, carteolol hydrochloride, carvedilol, celiprolol hydrochloride, esmolol hydrochloride, isoprenaline hydrochloride, labetalol hydrochloride, metoprolol, metoprolol tartrate, nadolol, nebivolol hydrochloride, oxprenolol hydrochloride, penbutolol, pindolol, practolol, propranolol hydrochloride, sotalol hydrochloride, and timolol maleate) were included. Drugs other than β -blockers cohort included individual treatment with CCBs, angiotensin-converting-enzyme inhibitors, diuretics, α -blockers, aldosterone receptor antagonists, and digitalis.

Exposure time of drug was calculated from the time between the first prescription after the index date and the last prescription recorded in CPRD. The discontinuation date was defined as the last therapy prescription date prior to having a gap of at least 90 days after the last prescription was scheduled to expire.

2.4. Study outcomes

The primary outcome was total mortality, defined as date of death by any cause recorded in the CPRD. Secondary CV outcomes were nonfatal MI, nonfatal stroke, nonfatal arrhythmia, an embolism event and an angina episode, all defined by READ codes. Other secondary outcomes include mean proportional change of diastolic BP (DBP), systolic BP (SBP), and heart rate.

2.5. Covariates

Within each treatment cohort, data were extracted on patient age, sex, body mass index (BMI), heart rate, DBP, SBP, serum creatinine, smoking status, comorbidities, and medications at baseline. Comorbidities of interest were MI, chronic heart failure, asthma, chronic obstructive pulmonary disease (COPD), diabetes, peripheral vascular disease, atrial fibrillation, interstitial pulmonary disease, erectile dysfunction, and dyslipidemia. Information on current treatments collected at baseline included use of aspirin, statins, as well as previous coronary artery bypass grafting or surgery. Adherence to treatment, ie, the extent to which a patient acts in accordance with the prescribed interval, and dose of a dosing regime, was calculated from the number of days of medication supplied within a refill interval in relation to the number of days in the refill interval.

2.6. Data analysis

Descriptive statistics of baseline and treatment characteristics were performed according to treatment cohort, at index. Categorical variables were summarized using frequencies, percentages, and missing values. Continuous variables were summarized using mean, standard deviation. Adherence was calculated using the following formula: Total days supply / (last prescription date – first prescription date + last prescription day of supply) and reported as a proportion, with a cut-off of 80% or above defined as being adherent [22]. To illustrate the effect over time, the analysis of the outcomes was performed using cumulative follow-up times, categorized as < 1 year; < 2 years, < 5 years, < 10 years, and total (≤ 14 years). Patients surviving beyond the thresholds were right censored. Crude incidence rates (IR) for each treatment cohort were calculated as the number of each outcome per 1000 person-years (PY) with corresponding 95% confidence intervals (CI) for each reporting interval. Cox proportional hazard models were conducted to estimate the crude and adjusted hazard ratio (HR) by age and sex of each outcome in head-to-head comparison between treatment cohorts (bisoprolol vs other β -blockers; and bisoprolol vs drugs

Table 1
Patients and treatment characteristics.

	Bisoprolol (n = 987)	Other β -blocker (n = 1348)	Drugs other than β -blocker (n = 5272)
Age, years, mean (SD)	63.1 (11.5)	62.5 (11.6)	64.6 (12.5)
18 to 49, n (%)	121 (12.3)	176 (13.1)	601 (11.4)
50 to 69, n (%)	586 (59.3)	802 (59.5)	2823 (53.5)
≥ 70 , n (%)	280 (28.4)	370 (27.4)	1848 (35.1)
Sex, male, n (%)	639 (64.7)	867 (64.3)	3397 (64.4)
BMI, kg/m ² , mean $\pm$ SD n (%) †	27.1 $\pm$ 3.8 119 (12.1)	26.3 $\pm$ 3.5 204 (15.1)	26.7 $\pm$ 3.8 805 (15.3)
Heart rate/min†, mean $\pm$ SD	75.6 $\pm$ 9.0	76.8 $\pm$ 9.7	76.2 $\pm$ 12.1
≥ 70 /min, n (%)†	29 (2.9)	21 (1.6)	101 (1.9)
DBP, mm Hg, mean $\pm$ SD n (%) †	78.6 $\pm$ 7.9 132 (13.4)	82.0 $\pm$ 10.0 254 (18.8)	81.8 $\pm$ 10.7 962 (18.3)
SBP, mm Hg, mean $\pm$ SD n (%) †	134 $\pm$ 14.6 132 (13.4)	139.7 $\pm$ 18.7 254 (18.8)	141.1 $\pm$ 18.9 963 (18.3)
Comorbidity, n (%)			
MI	24 (2.4)	51 (3.8)	517 (9.8)
Chronic heart failure	4 (0.4)	3 (0.2)	158 (3.0)
Asthma	94 (9.6)	81 (6.0)	838 (15.9)
Atrial fibrillation	21 (2.2)	21 (1.6)	295 (5.6)
Cerebrovascular disease	83 (8.4)	73 (5.4)	864 (16.4)
COPD	10 (1.0)	10 (0.7)	125 (2.4)
Diabetes	28 (2.8)	36 (2.7)	269 (5.1)
Dyslipidemia	29 (2.9)	14 (1.0)	341 (6.5)
Erectile dysfunction	41 (4.2)	53 (3.9)	437 (8.3)
Interstitial pulmonary disease	4 (0.4)	3 (0.2)	73 (1.4)
Comedications, n (%)†			
NSAID	193 (19.6)	199 (14.8)	1007 (19.1)
Anti-platelets	322 (32.6)	341 (25.3)	2086 (39.6)
Statins	71 (7.2)	68 (5.0)	453 (8.6)
Treatment duration, years, median (IQR)	1.5 (0.3, 3.6)	2.6 (0.5, 6.6)	2.5 (0.6, 5.9)
Treatment discontinued, n (%)	127 (12.9)	256 (19.0)	929 (17.6)
Adherence $\geq 80\%$, n (%)	894 (90.6)	1159 (85.9)	3331 (63.2)

BMI, body mass index; COPD, chronic obstructive pulmonary disease; DBP, diastolic blood pressure; IQR, interquartile range; MI, myocardial infarction; NSAID, nonsteroidal anti-inflammatory drug; SBP, systolic blood pressure; SD, standard deviation. †Measured at baseline.

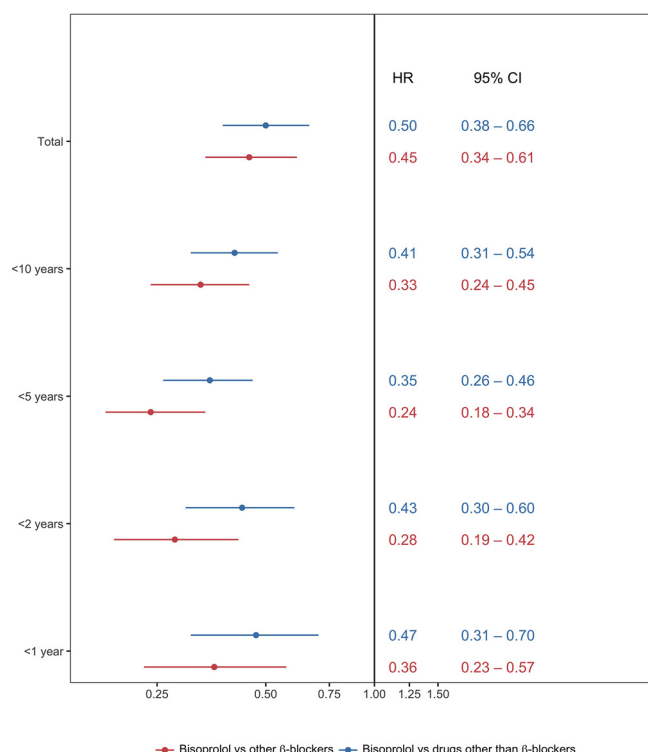


Fig. 1. Longitudinal mortality HR (95% CI) for bisoprolol versus other β -blockers and bisoprolol versus drugs other than β -blockers, adjusted for age and sex.

CI, confidence interval; HR, hazard ratio.

other than β -blockers). HR were presented with corresponding 95% CI. The survival probability of all-cause mortality was estimated using the Kaplan–Meier method. The statistical package used for all analyses was

SAS 9.4 (SAS Institute Inc., Cary, CA, USA).

3. Results

3.1. Patient characteristics

Of the 7607 patients included in the analysis, 987 (12.9%) were treated with bisoprolol, 1348 (17.7%) with other β -blockers and 5272 (69.4%) with drugs other than β -blockers. The baseline demographic and clinical characteristics of these patients are shown in Table 1. Patients had a mean age ranging from 63 to 65 years across the three different cohorts and around 65% were male regardless of the cohort. Cerebrovascular disease and asthma were the most common comorbidities in all three cohorts. The percentage of the patients receiving comedications at baseline was lower in those receiving other β -blockers compared with bisoprolol or drugs other than β -blockers. Mean BMI and heart rate were similar in all cohorts, but mean SBP and DBP were lower in the bisoprolol cohort (Table 1).

Treatment duration was lower in the bisoprolol cohort (median: 1.5 years, interquartile range [IQR]: 0.3–3.6) than in the cohorts receiving another β -blocker, or a drug other than a β -blocker (median: 2.6, IQR: 0.5–6.6 and median: 2.5, IQR: 0.6–5.9, years respectively). However, fewer patients discontinued treatment with bisoprolol (12.9%) compared with the other agents (19% in the β -blocker cohort and 17.6% in the drugs other than β -blocker cohort). A higher percentage of patients in the bisoprolol cohort (90.6%) were adherent (using $\geq 80\%$ as a cut-off), compared with patients receiving a different β -blocker (85.9%) or a drug other than a β -blocker (63.2%).

3.2. Mortality

After adjustment for age and sex, patients treated with bisoprolol had significantly improved survival over time compared with other β -blockers, and versus drugs other than β -blockers during all assessed time periods (Fig. 1). Overall, in the bisoprolol cohort, the risk of



Fig. 2. Kaplan–Meier survival curves for 14-year mortality.

mortality was decreased by 55% compared with patients who were treated with other β -blockers (HR: 0.45; 95% CI: 0.34–0.61) and 50% vs drugs other than β -blockers (HR: 0.50; 95% CI: 0.38–0.66). Likewise, the protective effect of bisoprolol in terms of mortality remained significant during the follow-up time (14 years), both when compared with other β -blockers and when compared with drugs other than β -blockers (Fig. 1). The Kaplan–Meier survival curves for total mortality with respect to treatment cohorts are presented in Fig. 2.

The IR for mortality was lowest in the bisoprolol cohort than in the other β -blocker cohort and the drugs other than β -blocker cohort at all time points (Fig. 3). Across all years, the mortality IR was 15.27 (95% CI: 11.47–19.92) deaths per 1000 PY in the bisoprolol cohort, 26.64 (95% CI: 23.26–30.37) deaths per 1000 PY in the other β -blockers cohort, and 31.80 (95% CI: 29.76–33.94) deaths per 1000 PY in drugs other than β -blocker cohort.

3.3. Cardiovascular outcomes

The adjusted HRs for secondary angina over the follow-up time (14 years) (HR: 0.58; 95% CI: 0.50–0.68) and MI < 5 years (HR: 0.45; 95% CI: 0.27–0.75) are significantly lower in patients with angina treated with bisoprolol versus patients treated with other β -blockers (Table 2). Adjusted HRs for arrhythmia and stroke showed no difference in risk (Table 2).

When compared with drugs other than β -blockers, adjusted HRs for bisoprolol treatment were significantly reduced for secondary angina (HR: 0.77; 95% CI: 0.68–0.88) and MI (HR: 0.34; 95% CI: 0.23–0.52) at all reporting intervals and arrhythmia < 5 years (HR: 0.61; 95% CI: 0.40–0.93). Stroke showed a decreased risk, although, it was only significant after 2 years (HR for < 5 years: 0.38; 95% CI: 0.17–0.82 and HR for 10 years: 0.45; 95% CI: 0.21–0.97).

In each treatment cohort, the incidence of CV events over time showed a downward trend. Longitudinal crude IRs for the cardiac outcomes showed a decreased incidence of angina and embolism at all reporting intervals for patients treated with bisoprolol, versus patients treated with other β -blockers, and patients treated with drugs other than β -blockers (Fig. 3). IRs for arrhythmia, MI, and stroke were also higher in patients treated with drugs other than β -blockers when compared with bisoprolol treatment at all reporting intervals.

4. Discussion

To our knowledge, this is the largest and longest duration study to evaluate the effects of bisoprolol on mortality and CV outcomes in patients after first diagnosis of angina under standard of care and monotherapies. These data provide real-world, long-term evidence to support previous meta-analytic findings that β -blockers have a beneficial effect on survival in patients with CAD [9].

4.1. Mortality outcomes

These data suggest that prescribing bisoprolol after the first episode of angina was associated with lower subsequent mortality relative to either treatment with other β -blockers, or drugs other than β -blockers, therefore, the primary outcome was met. Benefit was maintained over 14 years, bisoprolol was associated with significantly better survival and showed a reduction by 55% in the risk of mortality versus other β -blockers, and by 50% versus drugs other than β -blockers. Our results confirm the recommendation from current guidelines suggesting the use of β -blockers as the first-line agents in patients with stable angina.

4.2. Cardiovascular outcomes

In addition to improved survival, use of bisoprolol was associated with improved CV outcomes versus other β -blockers and drugs other than β -blockers. A significant benefit (over 14 years) in terms of secondary angina and MI was observed in bisoprolol cohort compared with other treatments cohorts. A significant reduction in the risk of arrhythmia was observed with bisoprolol treatment compared with other treatment. However, these data should be considered in the context that a greater proportion of patients receiving bisoprolol had a heart rate of at least 70 beats/min, a known risk factor for the development of ventricular arrhythmias in patients with coronary atherosclerosis [23].

4.3. Implications for guidelines

Despite the beneficial effect of bisoprolol on survival rates and other CV outcomes including MI, arrhythmia, and stroke, only 12.9% of patients received bisoprolol after the diagnosis of angina, suggesting that this β -blocker is underprescribed in comparison to other treatment options for this condition, despite guidelines relevant to the UK recommending β -blockers as first-line therapy [5,8].

This finding could be explained by some baseline differences between the three cohorts. The presence of severe health conditions and comorbidities at baseline might have affected the prescription pattern of bisoprolol. At baseline, patients receiving drugs other than β -blockers had a higher proportion of MI (2.4% vs 9.8%, respectively), cerebrovascular disease (8.4% vs 16.4%, respectively), diabetes (2.8% vs 5.1%, respectively), and dyslipidemia (2.9% vs 6.5%, respectively) compared with patients receiving bisoprolol.

The average duration of bisoprolol therapy in this study was shorter (1.5 years) than the duration of treatment in the other cohorts (2.6 and 2.5 years for other β -blockers and drugs other than β -blockers, respectively), yet patients receiving bisoprolol had a lower rate of discontinuation (12.9% in bisoprolol cohort compared with 19% in other β -blocker cohort and 18% in drugs other than β -blockers cohort). Overall, 90.6% of patients in the bisoprolol cohort were $\geq 80\%$ adherent with their medication compared with only 63.2% of patients in the cohort of patients receiving a drug other than β -blockers and 85.9%

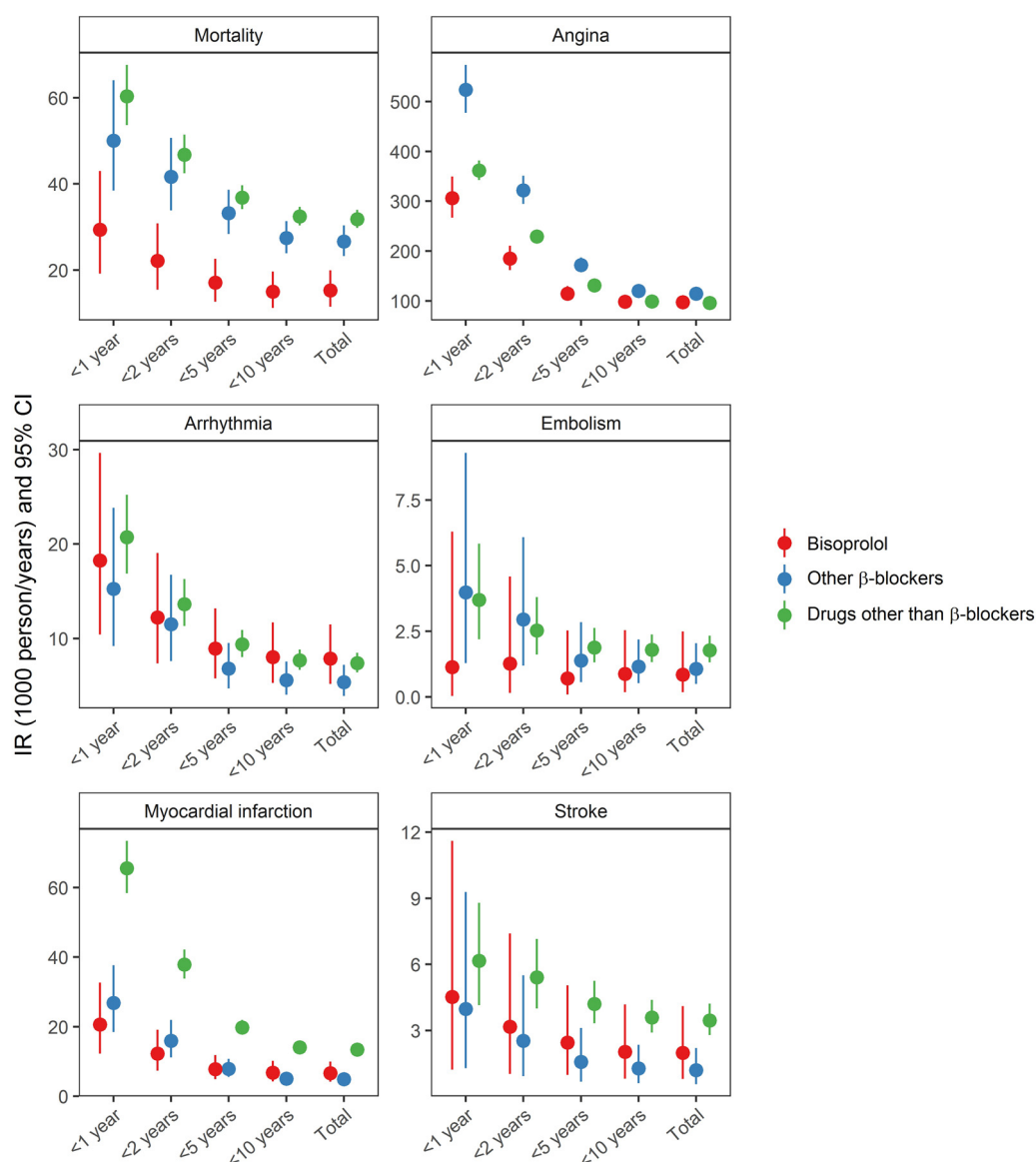


Fig. 3. Incidence rate per 1000 PY (95% CI) for mortality and each CV outcome over time. CI, confidence interval; CV, cardiovascular; IR, incidence rate; PY, person-years.

Table 2

Longitudinal HR with 95% CI of CV outcomes over time for bisoprolol versus other β-blockers and bisoprolol versus drugs other than β-blockers, adjusted for age and sex.

	Adjusted HR (95% CI)				
	< 1 year	< 2 years	< 5 years	< 10 years	Total (≤ 14 years)
Bisoprolol vs other β-blockers					
Arrhythmia	0.67 (0.34–1.32)	0.57 (0.32–1.03)	0.60 (0.36–1.00)	0.85 (0.52–1.38)	1.09 (0.67–1.78)
Embolism	0.16 (0.02–1.36)	0.20 (0.04–0.98)	0.23 (0.04–1.11)	0.46 (0.12–1.74)	0.60 (0.16–2.29)
Stroke	0.68 (0.18–2.55)	0.62 (0.19–2.04)	0.72 (0.26–1.98)	0.92 (0.35–2.44)	1.20 (0.45–3.20)
Angina	0.58 (0.49–0.68)	0.48 (0.41–0.56)	0.43 (0.37–0.50)	0.50 (0.43–0.58)	0.58 (0.50–0.68)
MI	0.44 (0.25–0.78)	0.39 (0.23–0.69)	0.45 (0.27–0.75)	0.66 (0.40–1.11)	0.84 (0.50–1.42)
Bisoprolol vs drugs other than β-blockers					
Arrhythmia	0.55 (0.32–0.94)	0.61 (0.37–0.99)	0.61 (0.40–0.93)	0.78 (0.52–1.16)	0.92 (0.61–1.37)
Embolism	0.18 (0.02–1.38)	0.33 (0.08–1.39)	0.23 (0.05–0.94)	0.37 (0.12–1.19)	0.45 (0.14–1.44)
Stroke	0.50 (0.17–1.45)	0.41 (0.16–1.04)	0.38 (0.17–0.82)	0.45 (0.21–0.97)	0.54 (0.25–1.17)
Angina	0.71 (0.61–0.88)	0.65 (0.56–0.74)	0.61 (0.54–0.70)	0.70 (0.62–0.80)	0.77 (0.68–0.88)
MI	0.19 (0.12–0.31)	0.20 (0.12–0.31)	0.22 (0.15–0.35)	0.30 (0.19–0.45)	0.34 (0.23–0.52)

CI, confidence interval; CV, cardiovascular; HR, hazard ratio; MI, myocardial infarction. Numbers in **bold** represent a reduction in risk.

of patients receiving other β -blockers, and this could explain the better outcomes in the bisoprolol cohort versus the cohort receiving drugs other than β -blockers. Studies have shown consistently better CV outcomes among patients with stable angina who are adherent with medical therapy, including β -blockers, compared with less adherent patients [24,25].

4.4. Strengths and limitations

The strength of the study includes long duration of follow-up and study population, which is representative of the UK primary care population.

There are several limitations to the study. The outcomes analysis in this study may have been confounded by differences in baseline characteristics and comorbidities between treatment cohorts. For example, patients who received drugs other than β -blockers included a higher proportion of patients aged ≥ 70 years, and a higher proportion of patients with comorbidities such as diabetes, COPD, cerebrovascular disease, prior MI, and dyslipidemia compared with the patients receiving bisoprolol or other β -blockers. As a result, patients in this cohort were also more likely to be receiving a range of concomitant medications. The analysis was not adjusted for baseline comorbidities for the patients in each treatment group. These differences in baseline characteristics might partially explain the improved mortality in the bisoprolol group. However, the Cox regression analysis was adjusted for age and sex, which are relevant confounders. Likewise, residual confounding by some variables (such as BP or heart rate, which were missing for most patients) may remain and might have affected the outcome. In addition, the bisoprolol cohort contained fewer patients with high DBP or SBP compared with the cohorts, suggesting that patients in the bisoprolol cohort were healthier overall. Also, the bisoprolol cohort contained a higher proportion of patients with elevated serum creatinine or heart rate of at least 70 beats/min.

Finally, the study was conducted in primary care patients in the UK, and may not be applicable to other patient populations.

5. Conclusions

These long-term data suggest that the treatment of new-onset angina with bisoprolol was associated with a significant reduction in the risk of mortality by at least 50%, and a number of CV outcomes compared with other treatments in a real-world primary care population. However, studies comparing bisoprolol side by side to other commonly used beta blockers are still needed. These results confirm the first-line recommendation of treatment of angina with bisoprolol instead of other β -blockers and drugs other than β -blockers. In the UK primary care setting, bisoprolol is rarely prescribed for patients with angina and given the protective effect in mortality in the long term, we propose that bisoprolol should be more widely utilized.

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Statement of authorship

This authors takes responsibility for all aspects of the reliability and freedom from bias of the data presented and their discussed interpretation.

Conflicts of interest

Guido Grassi declares lecturer payments from Merck KGaA.

Thilo Hohenberger is an employee of Merck KGaA.

Meritxell Sabidó is an employee of Merck KGaA.

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